Two Exact Counterexamples in Weighted Rounding
Illustrated Research Overview

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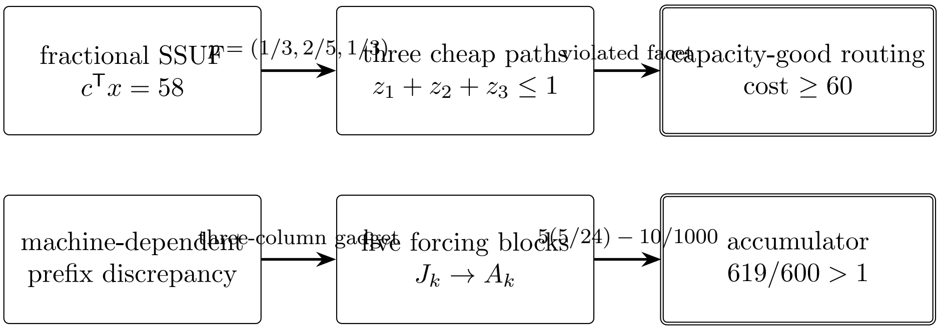
23 July 2026

Abstract

This note summarizes two exact certificate-based counterexamples: a planar seven-vertex counterexample to Goemans' cost conjecture for single-source unsplittable flow, and an 11-row, 15-column counterexample to the machine-dependent weighted-chairman conjecture. Both are accompanied by standard-library Python verifiers using exact arithmetic.



1 The flow result in one page

For demands $(15, 10, 15)$, the planar acyclic network in Figure 1 supports a feasible fractional flow of cost 58. Each terminal has one expensive path of total integral cost 30 and one zero-cost path. Any pair of zero-cost paths violates one of three arc bounds by exactly one unit. A capacity-good routing therefore buys at least two expensive paths and costs at least 60.

The convex-geometric obstruction is the stable-set inequality of a triangle:

$$z_1 + z_2 + z_3 \le 1, \qquad \frac{1}{3} + \frac{2}{5} + \frac{1}{3} = \frac{16}{15} > 1.$$

The example is planar, but its underlying undirected graph is a subdivision of K_4, immediately beyond the series-parallel class for which the exact positive theorem is known [1].

A parametric form gives the quantitative planar lower bound

$$\frac{9}{8} \le \alpha_{\mathrm{planar}} \le 2,$$

where $\alpha_{\mathrm{planar}} D$ is the permitted cost-preserving additive upper deviation and the upper bound is the planar theorem of Traub et al. [3].

2 The assignment result in one page

Liu and Reis conjectured prefix discrepancy at most $D = \max_{i,j} d_{ij}$ for arbitrary positive machine-dependent weights [2]. The counterexample repeats the detector shown in Figure 2 five times.

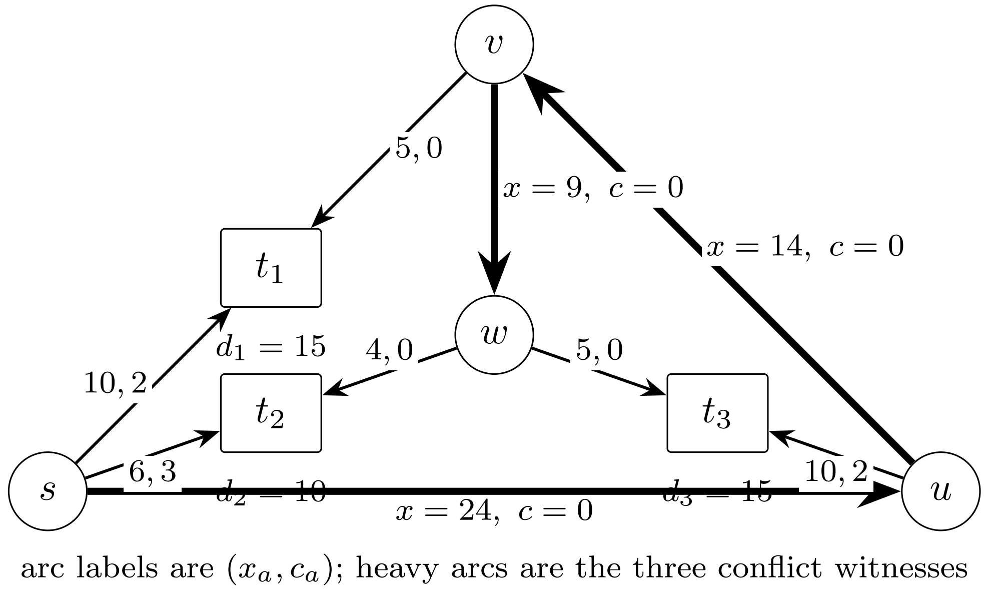


Figure 1: The explicit planar counterexample. Arc labels are (fractional load, per-unit cost).

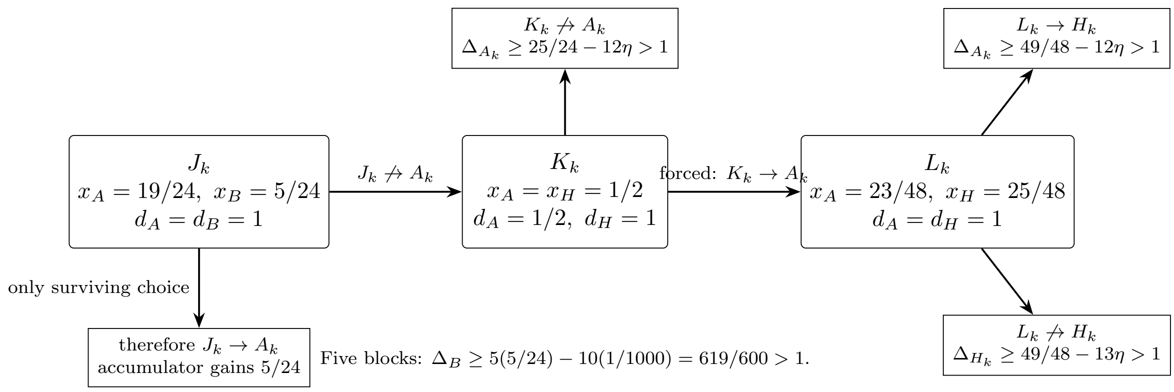


Figure 2: One forcing block. No support restriction is assumed.

Every discrepancy-good assignment is forced to send J_k to A_k. Each such choice contributes 5/24 to a global accumulator row. The ten detector columns can repair the accumulator by at most 10/1000, leaving

$$5\left(\frac{5}{24}\right) - \frac{10}{1000} = \frac{619}{600} > 1.$$

Thus the unrestricted machine-dependent conjecture and its stronger unified support-preserving version are false.

3 Why the results matter

The flow result preserves the classical Dinitz–Garg–Goemans theorem: the failure occurs only when cost preservation is imposed. The chairman result preserves the ordinary common-weight problem: the failure occurs only after weights become row dependent.

4 Verification

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Table 1: Conservative chronology and significance.

period	development	significance
1998–1999	Dinitz–Garg–Goemans additive-D theorem	Establishes optimal cost-free congestion rounding.
2000–2002	minimum-cost SSUF formulation	Goemans' simultaneous cost/congestion conjecture enters the literature.
2024	planar additive-$2D$ cost theorem	Best known planar upper bound before the present certificate.
2024–2025	exact series-parallel convex decomposition	Confirms the roughly 25-year-old conjecture on a nontrivial class.
2026	present seven-vertex certificate	Shows the exact D cost guarantee fails even for planar DAGs.
2025–2026	Liu–Reis machine-dependent conjecture	Natural extension of weighted chairman assignment.
2026	present five-block forcing certificate	Shows machine dependence destroys the unit prefix bound.

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python3 code/verify_all.py
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checks the flow graph, reconstructs all six source–terminal paths, enumerates all eight routings, and checks every rational inequality in the chairman forcing proof. No floating-point computation or optimization solver is needed.

Status. These are preprint claims prepared for independent review. Human authors must verify priority, theorem wording, and the final submission. A generative AI system assisted with exploration and drafting and is not an author.

References

[1] M. M. Almoghrabi, M. Skutella, and P. Warode. Integer and unsplittable multiflows in series-parallel digraphs, 2024.

[2] X. Liu and V. Reis. The weighted chairman assignment problem. In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, volume 362 of *Leibniz International Proceedings in Informatics*, pages 98:1–98:16. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi: 10.4230/LIPIcs.ITCS.2026.98.

[3] V. Traub, L. Vargas Koch, and R. Zenklusen. Single-source unsplittable flows in planar graphs. In *Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 639–668. SIAM, 2024. doi: 10.1137/1.9781611977912.24.